ABELIAN SHEAVES ON A TOPOLOGICAL SPACE HAVE FUNCTORIAL INJECTIVE EMBEDDINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete construction; the injectivity of the target is attributed to a named product lemma carried on this card as a navigation stub, and both asserted properties are otherwise verified in the span.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Functorial injective embeddings of abelian groups. Same chapter, the construction of a functorial embedding of an abelian group into an injective group. Supplies the embeddings of the groups that appear in the skyscraper construction.
- Skyscraper sheaves and their adjunction with the stalk functor. Sheaves, Lemma sheaves-lemma-stalk-skyscraper-adjoint. Supplies the adjunction that identifies the sheaf constructed from the injective groups at the points as a product of skyscraper sheaves.
- Products of injective objects. Homology, Lemma homology-lemma-product-injectives. Supplies that a product of injective objects is injective, which gives injectivity of the constructed sheaf.

Source reference. Original section 01D8: sources/stacks-more-algebra/pinned-original-source.tex, source lines 13617–13617. Navigation pointer invoked at proof line 623; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 006X: sources/stacks-sheaves/pinned-original-source.tex, source lines 814–814. Navigation pointer invoked at proof line 667; the counted span contains the pointer only and the cited result is not reproduced

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Source reference. Original lemma 0079: sources/stacks-sheaves/pinned-original-source.tex, source lines 1163–1163. Navigation pointer invoked at proof line 677; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 0099: sources/stacks-sheaves/pinned-original-source.tex, source lines 3487–3487. Navigation pointer invoked at proof line 689; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 009C: sources/stacks-sheaves/pinned-original-source.tex, source lines 3549–3549. Navigation pointer invoked at proof line 696; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0137: sources/stacks-homology/pinned-original-source.tex, source lines 7335–7335. Navigation pointer invoked at proof line 699; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let X be a topological space. The category of abelian sheaves on X has enough injectives. In fact it has functorial injective embeddings.*

Proof. For an abelian group A we denote $j : A \to J(A)$ the functorial injective embedding constructed in More on Algebra, Section 01D8. Let $\mathcal{F}$ be an abelian sheaf on X. By Sheaves, Example 006X the assignment

$$\mathcal{I} : U \mapsto \mathcal{I}(U) = \prod\nolimits_{x \in U} J(\mathcal{F}_x)$$

is an abelian sheaf. There is a canonical map $\mathcal{F} \to \mathcal{I}$ given by mapping $s \in \mathcal{F}(U)$ to $\prod_{x \in U} j(s_x)$ where $s_x \in \mathcal{F}_x$ denotes the germ of s at x. This map is injective, see Sheaves, Lemma 0079 for example.

It remains to prove the following: Given a rule $x \mapsto I_x$ which assigns to each point $x \in X$ an injective abelian group the sheaf $\mathcal{I} : U \mapsto \prod_{x \in U} I_x$ is injective. Note that

$$\mathcal{I} = \prod\nolimits_{x \in X} i_{x,*} I_x$$

is the product of the skyscraper sheaves $i_{x,*} I_x$ (see Sheaves, Section 0099 for notation.) We have

$$\mathrm{Mor}_{Ab}(\mathcal{F}_x, I_x) = \mathrm{Mor}_{Ab(X)}(\mathcal{F}, i_{x,*} I_x).$$

see Sheaves, Lemma 009C. Hence it is clear that each $i_{x,*} I_x$ is injective. Hence the injectivity of $\mathcal{I}$ follows from Homology, Lemma 0137. □